

CODE ONLY: Experiments 1 to 10

Yellow line = this line was wrong in the manual and has been corrected. All other lines are exactly as in your manual. Language (Scilab / MATLAB) is the same as the one your manual used for that experiment.

EXPERIMENT 01 · Vector Algebra: Components and Magnitude of a Vector (Scilab)

```
clc
clear
format('v',6)
A=[10,-4,6]
B=[2,1,0]
disp(A(1,2),'component of A along ay:')
P=3*A-B
disp((P(1,1)^2+P(1,2)^2+P(1,3)^2)^0.5,'magnitude is:')
c=A+2*B
det_c=(c(1,1)^2+c(1,2)^2+c(1,3)^2)^0.5
format('v',7)
ac=c/det_c
disp(ac,'unit vector along c is:')
```

Correction:

- $\mathbf{A} = [-10, -4, 6] \rightarrow [10, -4, 6]$ (A is $10\mathbf{a}_x - 4\mathbf{a}_y + 6\mathbf{a}_z$).
- $\mathbf{p} = 3\mathbf{A} - \mathbf{B} \rightarrow \mathbf{P} = 3\mathbf{A} - \mathbf{B}$ (Scilab is case-sensitive; P is used on the next line).
- $\mathbf{dot_c} \rightarrow \mathbf{det_c}$ (the variable name must be the same where it is defined and used).

EXPERIMENT 02 · Vector Algebra: Dot Product and Cross Product (Scilab)

```
clc
clear
A=[3,4,1]
B=[0,2,-5]
det_A=(A(1,1)^2+A(1,2)^2+A(1,3)^2)^0.5
det_B=(B(1,1)^2+B(1,2)^2+B(1,3)^2)^0.5
theta=acosd((sum(A.*B))/(det_A*det_B))
disp(theta,'Angle b/w A and B is')
```

Correction:

- $\mathbf{sum(A*B)} \rightarrow \mathbf{sum(A.*B)}$. Two row vectors cannot be multiplied with $*$; use $.*$ (element-wise), then $\mathbf{sum} = \mathbf{dot}$ product.

EXPERIMENT 03 · Transmission Line: Finding Various Parameters (Scilab)

```
clear
clc
Z0=60, a=20*10^-3, u=0.6*3*10^8, f=100*10^6
R=a*Z0, disp(R,'R=')
L=Z0/u, disp(L*10^9,'L in nH=')
G=a*a/R, disp(G*10^6,'G in micro S per meter=')
C=1/(u*Z0), disp(C*10^12,'C in pf=')
l=u/f, disp(l,'l=')
```

Correction:

- First line: $\mathbf{a=20*10^3} \rightarrow \mathbf{20*10^{-3}}$ ($\alpha = 20 \text{ mNp/m} = 0.02$) and $\mathbf{u=6*3*10^8} \rightarrow \mathbf{0.6*3*10^8}$ ($u = 0.6c = 1.8 \times 10^8$). Your own output ($a = 0.02$, $u = 1.800\text{D}+08$) confirms these values. Rest of the code is correct as written.

EXPERIMENT 04 · Coulomb's Law (Scilab)

```
clear
clc
format('v',6)
Q1=1
Q2=-2
Q=10*10^-9
P1=[0 3 1]-[3 2 -1]
P2=[0 3 1]-[-1 -1 4]
e=10^-9/(36*pi)
det1=(P1(1,1)^2+P1(1,2)^2+P1(1,3)^2)^.5
det2=(P2(1,1)^2+P2(1,2)^2+P2(1,3)^2)^.5
F=((Q*Q1)*P1)/(4*pi*e*det1^3)+((Q*Q2)*P2)/(4*pi*e*det2^3)
E=[(10^-6)*(F/Q)]
disp(F,'F(in mN)=')
disp(E,'At that point E(in kV)=')
```

Correction:

- Line for **F**: the brackets in the manual are not balanced ($[(Q*Q1)*(P1)] \dots$). Rewritten with matching round brackets; the formula is unchanged.
- ($Q1 = 1$ and $Q2 = -2$ are in mC, so F comes out in mN; this is fine.)

EXPERIMENT 05 · Stokes' Theorem (Scilab)

```
clear
clc
ab=integrate('2*sin(P)', 'P', %pi/3, %pi/6);
bc=(3^.5/2)*integrate('P', 'P', 2, 5)
cd=integrate('5*sin(P)', 'P', %pi/6, %pi/3)
da=.5*integrate('P', 'P', 5, 2)
C1=ab+bc+cd+da
disp(C1, 'C1=')
C2=integrate('sin(P)', 'P', %pi/6, %pi/3)*integrate('(1+P)', 'P', 2, 5)
disp(C2, 'C2=')
disp('since C1=C2 hence stokes theorem is proved')
```

Correction:

- **C2 line**: integrate() needs the variable name: integrate('(1+P)', '**P**', 2, 5) (it was missing in the manual); also use the same plain variable P instead of the symbol Φ .
- **Last disp**: the apostrophe in stoke's ends the string early and gives an error; removed it.

EXPERIMENT 06 · Coordinate Systems and Transformation (MATLAB)

```
clc
clear all
close all
format
[x,y,z]=spto_cart(-3,4,0);
disp('B in cartesian co-ordinate :')
disp([x y z])
[r,e,z]=spto_cylin(5,pi/2,-2);
disp('B in cylindrical co-ordinate :')
disp([r e z])
function [x,y,z]=spto_cart(x,y,z)
R=sqrt(x^2+y^2+z^2);
r=sqrt(x^2+y^2);
P=asin(r/R);
Q=acos(x/r);
x=(10/R)*sin(P)*cos(Q)+R*(cos(P))^2*cos(Q)-sin(Q);
y=(10/R)*sin(P)*sin(Q)+R*(cos(P))^2*sin(Q)+cos(Q);
z=(10/R)*cos(P)-R*cos(P)*sin(P);
end
function [r,e,z]=spto_cylin(r,P1,z1)
R=sqrt(r^2+z1^2);
P=acos(z1/R);
r=(10/R)*sin(P)+R*(cos(P))^2;
z=(10/R)*cos(P)-R*cos(P)*sin(P);
e=1;
end
```

Correction:

- Function names cannot contain a space: **spto cart** → **spto_cart**, **spto cylin** → **spto_cylin** (in the header lines and in the calls).
- **R** is the spherical radius, so it must include z^2 : $R = \sqrt{x^2+y^2+z^2}$ (manual had only x^2+y^2 , which is ρ).
- Cylindrical part: the function returned **P1 (= $\pi/2$, the ϕ coordinate)**, which is why the manual output shows 1.5708. B_ϕ is 1, so it now returns **e=1**. Expected output: 2.4669 1.0000 1.1673.
- In MATLAB the functions must be at the end of the script file (as written here).

EXPERIMENT 07 · Different Components of a Vector (Scilab)

```
clc
clear all
close all
format('v',6)
function [R,P,Q]=Posvec(r,P,q)
R=r*sind(q); P=-sind(P)*cosd(q)/r; Q=r*r;
D=[R P Q]
disp(D,'D at P');
Dn=[r*sind(q) 0 0];
Dt=D-Dn;
disp(Dt,'Tangential component of D at P');
endfunction
Posvec(10,150,330)
D=[-5 -0.043 100];
q=[0 1 0]
U1=D(1,2)*q(1,3)-D(1,3)*q(1,2)
U2=D(1,3)*q(1,1)-D(1,1)*q(1,3)
U3=D(1,1)*q(1,2)-D(1,2)*q(1,1)
U=[U1 U2 U3]
det_U=sqrt(U1^2+U2^2+U3^2);
format('v',7);
disp(U/det_U,'Unit vector P perpendicular to D');
```

Correction:

- $U3^3 \rightarrow U3^2$ in $\det_U$ (magnitude = $\sqrt{U1^2 + U2^2 + U3^2}$).
- $D=[-5 \ -0.043 \ 100]$: D_θ is negative (-0.043), as in your output.
- Scilab closes a function with **endfunction** (not “End function”).

EXPERIMENT 08 · Dipole (Scilab)

```
clear
clc
P1=-5*10^-9, P2=9*10^-9
r1=2, r2=-3, e=10^-9/(36*pi);
v=(1/(4*pi*e))*((P1*abs(r1)/r1^3)+(P2*abs(r2)/r2^3));
disp(v);
```

Correction:

- $r2=3 \rightarrow r2=-3$. The dipole 2 is at $z = 3$, so the vector from it to the origin is $-3a_z$. With $r2 = 3$ the answer comes out -2.25 V instead of the correct -20.25 V (your output page also shows $r2 = -3$).

EXPERIMENT 09 · Electromagnetic Wave Propagation: Waves in Dielectric (MATLAB)

```
clear
clc
B=1
n=60*pi
Ur=1
E0=10^-9/(36*pi)
U0=4*pi*10^-7
Er=(U0*Ur)/(n^2*E0)
fprintf('Er = %.0f \n', Er)
w=B/sqrt(E0*Er*U0*Ur);
fprintf('w in Mrad/sec = %.0f \n', w*10^-6)
```

Correction:

- New-line must be written as `\n` (backslash), not `/n`.
- Last line: $w*10^{-6} \rightarrow w*10^{-6}$ (1 to any power is 1, so it would not convert to Mrad/s). Output: $Er = 4$, $w = 150 \text{ Mrad/s}$.

EXPERIMENT 10 · Waveguide: Wave Propagation in Guide (Scilab)

```
clear
clc
a=8.636*10^-2, b=4.318*10^-2, f=4*10^9
u=3*10^8
fc=u/(2*a)
disp(fc*10^-9,'cutoff frequency =')
if (f>=fc) then disp('As f>fc so TE10 mode will propagate')
else disp('It will not propagate')
end
Up=u/sqrt(1-(fc/f)^2)
disp(Up*10^-6,'Phase velocity in Mm/sec =')
Ug=u*u/Up
disp(Ug*10^-6,'Group velocity in Mm/sec =')
```

No correction needed; the code is correct as written in the manual.